

AI Engineer Guide (Improved Structure)

1. Who is an AI Engineer?

An AI Engineer designs, builds, deploys, and maintains AI-powered systems. The role focuses on applying machine learning and large language models to solve real-world business problems using sound software engineering practices.

2. Responsibilities

- Prepare and process data
- Train, evaluate, and improve models
- Build AI-powered APIs and applications
- Deploy models to production
- Monitor reliability, cost, and performance
- Collaborate with product, data, and engineering teams

3. Required Skills

Programming

- Python
- SQL

Machine Learning

- Scikit-learn
- PyTorch
- TensorFlow fundamentals

LLMs

- Prompt Engineering
- Embeddings
- Retrieval-Augmented Generation (RAG)
- Tool Calling

Backend

- FastAPI
- REST APIs

DevOps & MLOps

- Git
- Docker
- Kubernetes (preferred)
- CI/CD basics

Cloud

- AWS, Azure, or Google Cloud

4. Preferred Skills

Vector databases, MLflow, Weights & Biases, distributed systems, security, monitoring, optimization, and system design.

5. Programming Languages

Python, SQL, Bash, JavaScript/TypeScript, with optional exposure to Java, Go, or C++.

6. Frameworks

PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, FastAPI, LangChain/LlamaIndex.

7. Tools

VS Code, Jupyter, GitHub, Docker, PostgreSQL, FAISS/Chroma/Pinecone, cloud services, OpenAI-compatible APIs.

8. Learning Roadmap

Phase 1 – Foundations

Python → Git → SQL

Phase 2 – Data

NumPy → Pandas → Statistics

Phase 3 – Machine Learning

Scikit-learn → Model Evaluation → Feature Engineering

Phase 4 – Deep Learning

PyTorch → TensorFlow → Transformers

Phase 5 – LLM Engineering

Prompt Engineering → Embeddings → RAG → Agents

Phase 6 – Deployment

FastAPI → Docker → Cloud → Monitoring → MLOps

9. Beginner Projects

Spam classifier, sentiment analysis, house price prediction API, movie recommender, simple chatbot.

10. Intermediate Projects

Image classifier, document Q&A; with RAG, recommendation engine, OCR pipeline, model inference API.

11. Advanced Projects

Production AI assistant, multi-agent system, fine-tuned LLM, multimodal application, scalable MLOps pipeline.

12. Resume Optimization Tips

Emphasize measurable impact, deployed projects, GitHub portfolio, categorized technical skills, concise achievements, and clean formatting.

13. Interview Topics

Python, SQL, ML algorithms, evaluation metrics, neural networks, transformers, RAG, embeddings, APIs, Docker, deployment, system design, and behavioral questions.

14. Career Progression

Typical path: Student → Junior AI Engineer → AI Engineer → Senior AI Engineer → AI/ML Lead → AI Architect. Focus on progressively larger production systems, stronger engineering practices, and measurable business impact.